Library Requirements Specification

Version 3.0

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Use this Requirements Specification template to document the requirements for your product or service, including priority and approval. Tailor the specification to suit your project, organizing the applicable sections in a way that works best, and use the checklist to record the decisions about what is applicable and what isn't.

The format of the requirements depends on what works best for your project.

This document contains instructions and examples which are for the benefit of the person writing the document and should be removed before the document is finalized.

To regenerate the TOC, select all (CTL-A) and press F9.

Table of Contents

Library Managment System Requirements Specification

1. Executive Summary

1.1 Project Overview

This is a university project developed to facilitate and enhance access to library resources within an academic environment. An efficient solution is to digitalize the Library Management System, bringing it closer to all university members while providing better management tools for staff and administrators. By transforming traditional library processes into a digital platform, the system aims to improve organization, accessibility, and efficiency in managing library services.

This web application allows university members to search for books, borrow and return them, and pay overdue fines through a simple interface. Staff members manage the daily operations of the library, including finances, inventory, and facility maintenance, ensuring that the system operates smoothly.

Members can also leave reviews and feedback on books they have read, helping other users discover useful resources and allowing the library to better understand user preferences.

All users access the system through role-based permissions, ensuring that each user only interacts with the features related to their responsibilities.

Our work consists of taking the necessary steps that will lead to the creation and maintenance of a fully functional university library web application that improves both library management and user access to resources.

2. Product/Service Description

This section describes general factors that influence the Library Management System and its requirements. The system must support different user roles, manage books and borrowing transactions, handle fines, and maintain accurate records. It is designed to run in a web-based environment and help library staff perform daily operations efficiently. These factors explain why the system has the features specified in later requirements.

2.1 Product Context

How does this product relate to other products?

The Library Management System interacts with several supporting systems to perform its functions.

- It connects to a database system to store and manage information about books, users, borrowing records, and payments.
- It connects to an email or notification service to send reminders about due dates and overdue books.
- Users access the system through web browsers on computers or mobile devices.

Is it independent or self-contained?

The system is mostly self-contained, but it still depends on some external services.

- Most library operations (borrowing books, managing users, tracking inventory) happen inside the system.
- The system depends on a **database** to store and retrieve information.
- It requires an **internet connection** for users to access the system.
- It relies on external services such as **email notifications**.

Does it interface with a variety of related systems?

Yes, the system interfaces with several related systems.

- **Database Management System** – stores all system data.
- **Email / Notification Service** – sends alerts about due dates or overdue books.
- **Web Browser Interface** – allows users to access the system online.

Describe these relationships or use a diagram to show the major components of the larger system, interconnections, and external interfaces.

Major Components of the System

The system is composed of several main components.

- **User Interface** – the front-end where users interact with the system.
- **Application Logic** – handles operations such as borrowing books, issuing fines, and managing users.
- **Database** – stores all records related to books, users, transactions, and payments.

Internal Interconnections

- **User Interface ↔ Application Logic**

Users perform actions (search, borrow, return) on the interface; the logic processes requests.

- **Application Logic ↔ Database**
 All operations (book search, borrowing, fines) require reading from or writing to the database.

- **Application Logic ↔ Notification Service**
 When a book is due or overdue, the logic module sends a message through the notification service.

- **Application Logic ↔ Role & Permission Module**
 Ensures each user (admin, librarian, student) can only access their authorized functions.

External Interfaces

- **Database Management System** → stores books, users, transactions, fines.
- **Email / Notification Service** → sends reminders or alerts to users.
- **Web Browser / Mobile Browser** → users access the LMS.
- **Network / Internet** → enables all online communications.

2.2 User Characteristics

1.User (Student / Academic Staff using inheritance). Student and Academic Staff will inherit from the user. They interact with the platform to search for books, borrow and return materials, pay overdue fines, submit acquisition requests, and provide feedback. Students focus on accessing academic resources efficiently, while academic staff additionally require extended borrowing privileges and access to specialized materials for teaching and research purposes. Both groups interact with the system frequently and require a simple, intuitive interface.

2. Librarian - Librarians are trained staff responsible for daily system operations. They have intermediate to advanced system knowledge and use the platform to manage book circulation, update catalog records, and assist users. Their interaction with the system is continuous and operational.

3. Library Administrator - The administrator has advanced technical knowledge and full access to the system. They manage user roles, permissions, and system configurations, as well as monitor system performance and generate reports. Their usage is administrative and strategic.

4. Finance Officer - The finance officer interacts with the system to manage fines and payments. They require basic to intermediate technical skills and focus on tracking financial transactions, generating reports, and ensuring accurate handling of overdue fees.

5. Inventory Manager - The inventory manager is responsible for maintaining the physical collection of the library. They use the system to track book copies, update stock levels, and manage acquisitions or removals. Their interaction is task-oriented and requires attention to detail.

6. Maintenance Staff - Maintenance staff have minimal interaction with the system and basic technical skills. They use it only when necessary to report or track facility-related issues, ensuring that the physical library environment remains functional.

2.3 Assumptions

Internet Access

It is assumed that users (librarian, manager, and administrator) will have access to a stable internet connection in order to use the web-based system.

Device Availability

It is assumed that users will access the system through a computer or device with a modern web browser such as Chrome, Firefox, or Edge.

User Knowledge

It is assumed that users have basic computer skills, such as logging into a system, navigating web pages, and entering data into forms.

Database Availability

The system assumes that a database server is available to store information about books, employees, transactions, and stock.

User Authentication

It is assumed that each user (librarian, manager, administrator) will have a valid username and password provided by the administrator.

Data Accuracy

The system assumes that users will enter correct and accurate information when adding books, creating bills, or managing stock.

System Environment

It is assumed that the system will run in a standard web environment with necessary software such as a web server and database management system installed.

Access Control

It is assumed that users will only access features that correspond to their assigned roles

(Librarian, Manager, Administrator).

2.4 Constraints and Dependencies

The development and operation of the Library Management System are influenced by several technical, organizational, and operational constraints. These constraints define the boundaries within which the system must operate and identify dependencies on external components or processes.

System Platform Constraint
The system will be developed as a **web-based application**, meaning that users will access it through standard web browsers. This limits the system to technologies compatible with web environments and requires proper browser compatibility across commonly used browsers such as Chrome, Firefox, and Edge.

Database Dependency
The system depends on a **Database Management System (DBMS)** to store and manage all system data, including books, users, borrowing records, fines, and payment transactions. The system cannot operate without reliable database connectivity.

Network Dependency
Because the system is accessed online, it requires a **stable internet connection** for both library staff and members. Interruptions in network connectivity may temporarily prevent users from accessing the system or performing transactions.

Authentication and Role Management Constraint
All users must authenticate using valid credentials before accessing the system. Access to features is restricted by **role-based permissions** (e.g., student, librarian, administrator). This constraint ensures that users can only perform actions relevant to their responsibilities.

Integration with Notification Services
The system depends on an **external email or notification service** to send reminders about due dates, overdue books, and payment confirmations. This service must be available for the notification functionality to work correctly.

Policy and Operational Constraints
Library policies define certain operational rules within the system. For example, the default borrowing period is **7 days**, and extensions may only be granted under specific conditions. These policies must be enforced by the system.

Module Development Dependencies
Certain modules of the system depend on the completion of others. For example:

- The **user authentication module** must be implemented before other modules can verify user permissions.
- The **database structure** must be defined before implementing borrowing, payment, and inventory management features.

Data Integrity and Audit Constraints
The system must maintain accurate records of all transactions, including borrowing activities,

returns, payments, and administrative actions. Log files and activity tracking must be maintained for auditing and monitoring purposes.

3. Requirements

The Library Management System (LMS) is a web-based application designed to digitalize and streamline library services within a university environment. The system enables university members to search for books, borrow and return items, pay overdue fines, and submit reviews. Library staff and administrators use the system to manage inventory, finances, maintenance tasks, and operational processes. The system enforces **role-based permissions**, ensuring users access only the features relevant to their responsibilities.

- System Inputs

Input	Description
User login credentials	Username and password entered during login
Book search queries	Text input used to search library catalog
Borrow requests	Request submitted by members to borrow a book
Return records	Book return data entered by staff
Payment information	Data required to process overdue fine payments
Reviews and ratings	User-submitted book feedback
Inventory updates	Data entered by staff when managing book catalog

- System Outputs

Output	Description
Search results	List of books matching search criteria
Book details page	Complete information about a selected book

Borrow confirmation	Confirmation message when a book is borrowed
Return confirmation	Confirmation message when a book is returned
Fine notifications	Notifications about overdue books and fines
Payment receipts	Confirmation of successful fine payment
Reports	Financial and inventory reports generated for staff

Priority Definitions

The following definitions are intended as a guideline to prioritize requirements.

- Priority 1 – The requirement is a “must have” as outlined by policy/law
- Priority 2 – The requirement is needed for improved processing, and the fulfillment of the requirement will create immediate benefits
- Priority 3 – The requirement is a “nice to have” which may include new functionality It may be helpful to phrase the requirement in terms of its priority, e.g., "The value of the employee status sent to DIS **must be** either A or I" or "It **would be nice** if the application warned the user that the expiration date was 3 business days away". Another approach would be to group requirements by priority category.

- A good requirement is:
  - Correct
  - Unambiguous (all statements have exactly one interpretation)
  - Complete (where TBDs are absolutely necessary, document why the information is unknown, who is responsible for resolution, and the deadline)
  - Consistent
  - Ranked for importance and/or stability
  - Verifiable (avoid soft descriptions like “works well”, “is user friendly”; use concrete terms and specify measurable quantities)
  - Modifiable (evolve the Requirements Specification only via a formal change process, preserving a complete audit trail of changes)
  - Does not specify any particular design
  - Traceable (cross-reference with source documents and spawned documents).

3.1 Functional Requirements

In the example below, the requirement numbering has a scheme - BR_LR_0## (BR for Business Requirement, LR for Labor Relations). For small projects simply BR-## would suffice.

<actor>+ shall be able to do <operation> because < reason>

The following table is an example format for requirements. Choose whatever format works best for your project.

Req#	Requirement	Comments	Priority	Date	Rvwd SME / Approved
BR_LMS_01	The system must maintain a digital catalog that contains a record for every book and material owned by the library.	Business Process = Catalog Representation	1	3/14/2026	Sara Mjeshtri
BR_LMS_02	The system must assign a unique identifier to each book in the catalog.	Business Process = Resource Identification	1	3/14/2026	Sara Mjeshtri
BR_LMS_03	The system must organize books in the catalog according to a standardized classification system.	Business Process = Classification Standards	1	3/14/2026	Sara Mjeshtri
BR_LMS_04	The system must allow library staff to update the metadata of an existing catalog entry (e.g. title, author, ISBN, or classification).	Business Process = Catalog Maintenance	1	3/14/2026	Sara Mjeshtri
BR_LMS_05	The system shall store different editions of the same book as separate catalog entries.	Business Process = Edition Management	2	3/14/2026	Sara Mjeshtri

BR_LMS_06	The system shall include e-books in the catalog alongside physical books.	Business Process = Digital Resources	2	Sara Mjeshtri	Sara Mjeshtri
BR_LMS_07	The system shall allow users to search for books in the catalog using keywords.	Business Process = Catalog Search	2	3/14/2026	Sara Mjeshtri
BR_LMS_08	The system should support user account registration and creation.	Users must be able to create accounts to access library services.	1	3/14/2026	Klea Çela
BR_LMS_09	The system shall allow administrators to manage user accounts and assign roles..	Different roles require different permissions. Includes activation, deactivation, and modification of accounts.	1	3/14/2026	Klea Çela
BR_LMS_10	The system shall allow users to manage their profile information.	Users must be able to view and update their personal details.	2	3/14/2026	Klea Çela
BR_LMS_11	The system shall log all activities performed by actors for auditing and monitoring.	Activity logs help with security monitoring and auditing.	2	3/14/2026	Klea Çela
BR_LMS_12	The system shall manage library events, including creation, registration, and attendance tracking.	Covers creation and scheduling of events.	2	3/14/2026	Klea Çela

BR_LMS_13	The system shall allow the librarian to create and manage lost-and-found items and users to view lost-and-found items.	Librarian can post and manage lost-and-found items to help recover lost belongings..Users can view lost-and-found items.	1	3/14/2026	Klea Çela
BR_LMS_14	The system must allow librarians to record newly acquired books by entering detailed metadata	New Book Acquisition Recording	1	3/14/2026	Jorida Lilollari
BR_LMS_15	The system must maintain an accurate count of all physical copies available for each book title	Copy Count Tracking per Title	1	3/14/2026	Jorida Lilollari
BR_LMS_16	The system must allow librarians to mark specific copies of a book as damaged or lost and remove them from active circulation	Marking/Removing Damaged or Lost Copies	1	3/14/2026	Jorida Lilollari
BR_LMS_17	The system must automatically update the inventory and availability status of book copies whenever a borrowing or return transaction occurs	Automated Inventory Sync on Borrow or Return	1	3/14/2026	Jorida Lilollari
BR_LMS_18	The system must track the status of each individual physical copy (e.g., Available, Borrowed, Reserved, Damaged, Lost)	Status Tracking per Physical Copy	1	3/14/2026	Jorida Lilollari

BR_LMS_19	The system shall provide a structured process for users or staff to submit book acquisition requests, including request details, justification, and approval workflow	Formal Acquisition Request Process	2	3/14/2026	Jorida Lilollari
BR_LMS_20	The system shall generate detailed inventory reports for management, including total books, available copies, borrowed items, damaged/lost materials, and acquisition trends	Inventory Report Generation for Management	2	3/14/2026	Jorida Lilollari
BR_LMS_21	The system shall enforce loan duration policies based on user type (e.g., student, faculty, staff) and material type (e.g., books, journals, reference materials).	Business Process = "Borrowing Management". Different user categories have different borrowing periods. Reference materials may be in-library use only.	3	3/14/2026	Sibora Xhaferllari
BR_LMS_22	The system shall define and enforce borrowing limits per user, including maximum number of items allowed and maximum overdue items before restricting further borrowing.	Business Process = "Borrowing Management". Limits may vary by user type (e.g., undergraduates: 10 items, graduates: 15, faculty: 25).	3	3/14/2026	Sibora Xhaferllari

BR_LMS_23	The system shall process item returns, automatically update item status to "Available," and clear the loan record from the user's active borrowings.	Business Process = "Return Management". System updates inventory status in real-time and removes item from user's current loans.	3	3/14/2026	Sibora Xhaferllari
BR_LMS_24	The system shall track and maintain the status of each loan throughout its lifecycle (e.g., Active, Overdue, Returned, Renewed, Lost).	Business Process = "Loan Status Tracking". Status changes should be timestamped for audit purposes.	2	3/14/2026	Sibora Xhaferllari
BR_LMS_25	The system shall maintain a complete circulation history for each item and each user, including all borrowing and return transactions.	Business Process = "Circulation Records". Records needed for audit trails, dispute resolution, and usage analytics.	2	3/14/2026	Sibora Xhaferllari
BR_LMS_26	The system shall allow users to renew eligible items before or after the due date, with restrictions on renewal count based on item type and reservation status.	Business Process = "Borrowing Management". Items with pending reservations by other users should not be renewable.	2	3/14/2026	Sibora Xhaferllari
BR_LMS_27	The system shall send due date reminders to members before borrowed items become overdue.	Business Process = Due Date Reminder	2	3/14/2026	Jasmina Durmishi

BR_LMS_28	The system shall send overdue notifications to members when borrowed items are not returned on time.	Business Process = Overdue Notification	1	3/14/2026	Jasmina Durmishi
BR_LMS_29	The system shall send confirmation messages to members after a borrowing transaction is successfully completed.	Business Process = Borrowing Confirmation	2	3/14/2026	Jasmina Durmishi
BR_LMS_30	The system shall send confirmation messages to members after a return transaction is successfully completed.	Business Process = Return Confirmation	2	3/14/2026	Jasmina Durmishi
BR_LMS_31	The system shall notify members when fines are issued or updated on their account.	Business Process = Fine Notification	1	3/14/2026	Jasmina Durmishi
BR_LMS_32	The system shall deliver system alerts and library announcements to registered users.	Business Process = Alerts and Announcements	2	3/14/2026	Jasmina Durmishi
BR_LMS_33	The system must provide faculty with advanced filters to refine academic research results.	Business Process = Advanced Search. Traceable to Faculty profile.	1	3/14/2026	Arion Kanani
BR_LMS_34	The system shall enable Library Staff to moderate, hide, or pin user-generated reviews to maintain academic standards.	Business Process = Content Moderation. Traceable to Staff profile.	2	3/14/2026	Arion Kanani

BR_LMS_35	The system must generate read-only, encrypted financial reports for the Finance Officer regarding fine collections.	Business Process = Revenue Auditing. Traceable to Finance profile.	1	3/14/2026	Arion Kanani
BR_LMS_36	The system shall provide a technical bug-reporting interface that automatically attaches session logs for the System Admin.	Business Process = Bug Handling. Traceable to Admin profile.	1	3/14/2026	Arion Kanani
BR_LMS_37	The system must allow User (Student, Academic staff) to submit numerical ratings and written reviews for books.	Business Process = Peer Feedback	2	3/14/2026	Kristi Saliais
BR_LMS_38	The system must provide a dedicated feedback channel for User (Student, Academic staff) to report interface issues.	Business Process = System Improvement	1	3/14/2026	Kristi Saliasi
BR_LMS_39	The system shall provide personalized book recommendations for User (Student, Academic staff) based on history.	Business Process = User Engagement	3	3/14/2026	Kristi Saliasi

BR_LMS_40	The system must allow User (Student, Academic staff) to create and manage a personal "Wishlist" of books.	Business Process = Resource Tracking	2	3/14/2026	Kristi Saliasi
BR_LMS_41	The system must allow library staff to view and respond to User (Student, Academic staff) reviews and feedback.	Business Process = Staff Interaction	2	3/14/2026	Kristi Saliasi
BR_LMS_42	Inventory Manager purchases new book copies from an external supplier, from selecting titles issuing a Purchase Order, sending it to the supplier, receiving the shipment, verifying it, and closing the PO , so that the received copies can be cataloged and made available for loan.	Business Process= Purchase New Books	2	3/14/2026	Jorida Lilollari

3.2 Non-Functional Requirements

3.2.1 Product Requirements

Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

3.2.1.1 Usability Requirements

The Library Management System must provide a user-friendly and intuitive interface that allows both students and library staff to interact with the system efficiently without requiring advanced technical knowledge.

Ease of Learning (Learnability)
The system should be easy for first-time users to understand. New users should be able to learn basic operations such as searching for books, borrowing items, or managing inventory within a short period of use.

Clear and Intuitive Interface
The interface should present information in a clear and organized way. Menus, buttons, and navigation elements should be clearly labeled so that users can easily understand how to perform tasks such as searching for books, viewing account information, or processing returns.

Consistent Navigation
Navigation across different sections of the system should be consistent. Similar functions should follow the same interaction patterns so that users do not need to relearn the interface when performing different tasks.

Accessible Design
The system interface should be accessible from both **desktop and mobile devices** through standard web browsers. The design should remain usable on different screen sizes.

Error Prevention and Feedback
The system should help users avoid errors by validating inputs and providing clear feedback messages. For example, if a user enters incorrect login credentials or tries to borrow an unavailable book, the system should display an informative message explaining the issue.

Help and Documentation
The system should include help information or guidance for users. Basic documentation or tooltips should be available to explain key features such as borrowing procedures, payment of fines, and account management.

Efficiency of Use
Frequent users such as librarians should be able to perform common operations (e.g., registering returns, updating book records, generating reports) quickly with minimal steps.

3.2.1.2 Performance Requirements

This section specifies measurable requirements for the performance of the Library Management System.

- **Static numerical requirements:**
  - The system must support a minimum number of **200 concurrent users**.
  - It should handle up to **10,000 book records** in the catalog.
  - The system should support multiple user roles accessing the system simultaneously, including students, librarians, administrators, and finance staff.
- **Dynamic numerical requirements:**

- Search queries must return results within **2 seconds** under normal load.
- Borrowing and returning transactions should be processed within **3 seconds**.
- The system must maintain performance during peak periods, such as the start of a semester when many users borrow or return books.

3.2.1.3 Availability

Level of availability required

- The system **shall maintain at least 99% uptime during normal operating hours**.
- The system **shall be accessible to authorized users whenever the service is operational**.

Coverage for geographic areas

- The system **shall be accessible within the organization's network** (library/bookstore).
- The system **may also be accessed remotely through the internet by authorized users** with valid login credentials.

Impact of downtime on users and business operations

- If the system becomes unavailable, users **shall not be able to process transactions, update book inventory, or manage records**.
- Downtime **shall be minimized** to reduce interruptions to normal business operations.

Impact of scheduled and unscheduled maintenance

- The system **shall allow scheduled maintenance to be performed outside normal working hours whenever possible**.
- Users **shall be notified in advance** about planned maintenance periods.
- In the event of unscheduled maintenance, the system **shall be restored as quickly as possible** and users **shall be informed of service interruptions**.

Reliability

- The system **shall be designed to minimize system failures during normal operation**.

- The system **shall aim for a high Mean Time Between Failures (MTBF)**.
- The system **shall recover from failures quickly to restore normal functionality**.

3.2.1.4 Security

Encryption

- The system **shall encrypt sensitive information such as login credentials during transmission**.
- The system **shall protect stored sensitive data from unauthorized access**.

Activity logging and historical data sets

- The system **shall record logs of important user activities**, including logins, data modifications, and administrative actions.
- The system **shall store activity logs for auditing and monitoring purposes**.

Restrictions on intermodule communications

- The system **shall restrict communication between modules to authorized interactions only**.
- Each module **shall access only the data and functions necessary for its operation**.

Data integrity checks

- The system **shall validate user input before storing it in the database**.
- The system **shall prevent invalid, incomplete, or corrupted data from being stored**.
- The system **shall maintain consistency and accuracy of stored data**.

3.2.2 Organizational Requirements

Requirements which are a consequence of organisational policies and procedures e.g. process standards used, implementation requirements.

3.2.3 External Requirements

External requirements are conditions and constraints that come from factors outside the system and its development process. These requirements ensure that the system follows external rules, standards, and compatibility needs.

Interoperability Requirements

Even though the Library Management System is designed for a single university, it should still be able to interact with other internal university systems if needed. For example, it may connect with the student information system or authentication services to verify student identities and manage access permissions.

Legislative Requirements

The system must comply with data protection and privacy regulations when storing and managing user information. Personal data such as student names, identification numbers, and borrowing history must be securely stored and accessible only to authorized users.

Security Requirements

The system should provide secure access through authentication mechanisms such as usernames and passwords. In addition, role-based access control should be implemented so that different users (such as students, librarians, or administrators) can only access the features relevant to their roles.

Usability Requirements

The system should provide an intuitive and easy-to-use interface so that both students and library staff can interact with the platform without requiring advanced technical knowledge.

Platform Requirements

The system will be developed as a web-based application. Therefore, it must be accessible through standard web browsers such as Chrome, Firefox, or Edge, without requiring users to install additional software.

4. User Scenarios/Use Cases

Provide a summary of the major functions that the product will perform. Organize the functions to be understandable to the customer or a first time reader. Include use cases and business scenarios, or provide a link to a separate document (or documents). A business scenario:

● Describes a significant business need
● Identifies, documents, and ranks the problem that is driving the scenario
● Describes the business and technical environment that will resolve the problem ● States the desired objectives
● Shows the “Actors” and where they fit in the business model
● Is specific, and measurable, and uses clear metrics for success

Use cases are associated with a particular Functional Requirement. Assuming you have the first functional requirement named BR_01, you will map it into the Use Case called UC_01 and user scenario US_01. Please keep this naming convention throughout all your use cases and diagrams.

5. Diagrams

In this section you are going to place all of the diagrams that you build throughout to the course, in following with the slides presented throughout the weeks.

5.1 ER Diagram

Standard ERD for your project. Not much but the skills gained in the DBMS course are

required. 5.2 Use Case Diagram (general)

Use Case Diagram (only one, with all the use cases).

5.3 Activity Diagram

Each Activity Diagram should be associated with an use case, associated with a particular requirement which is further associated with a particular use-case. E.g BR_01 which becomes UC_01 which becomes AC_01.

5.4. Class diagram.

One class diagram (general) for all the classes. Edit it afterwards with the design pattern

implemented in it. 5.5 State diagram

Place all the relevant state diagrams here.

5.6 Sequence diagram.

All sequence diagrams are associated with an Activity Diagram. A Sequence Diagram is built based on an activity diagram. If the activity diagram is named AC_07, the Sequence Diagram will be named SC_07.

5.7. Collaboration diagram

All collaboration diagrams directly relate to a sequence diagram. If a sequence diagram is named SC_07, then the collaboration diagram is named CC_07

6. Design Patterns

Choose the relevant design patterns for your project. For each, give a reasoning and the associated class and sequence diagram. These are NOT part of the above diagrams, and need not carry the following naming scheme.

7. Appendix.

Organizing the Requirements

This section is for information only as an aid in preparing the requirements document.

Detailed requirements tend to be extensive. Give careful consideration to your organization scheme. Some examples of organization schemes are described below:

By System Mode

Some systems behave quite differently depending on the mode of operation. For example, a control system may have different sets of functions depending on its mode: training, normal, or emergency.

By User Class

Some systems provide different sets of functions to different classes of users. For example, an elevator control system presents different capabilities to passengers, maintenance workers, and fire fighters.

By Objects

Objects are real-world entities that have a counterpart within the system. For example, in a patient monitoring system, objects include patients, sensors, nurses, rooms, physicians, medicines, etc. Associated with each object is a set of attributes (of that object) and functions (performed by that object). These functions are also called services, methods, or processes. Note that sets of objects may share attributes and services. These are grouped together as classes.

By Feature

A feature is an externally desired service by the system that may require a sequence of inputs to affect the desired result. For example, in a telephone system, features include local call, call forwarding, and conference call. Each feature is generally described in a sequence of stimulus-response pairs, and may include validity checks on inputs, exact sequencing of operations, responses to abnormal situations, including error handling and recovery, effects of parameters, relationships of inputs to outputs, including input/output sequences and formulas for input to output.

By Stimulus

Some systems can be best organized by describing their functions in terms of stimuli. For example, the functions of an automatic aircraft landing system may be organized into sections for loss of power, wind shear, sudden change in roll, vertical velocity excessive, etc.

By Response

Some systems can be best organized by describing all the functions in support of the generation of a response. For example, the functions of a personnel system may be organized into sections corresponding to all functions associated with generating paychecks, all functions associated with generating a current list of employees, etc.

By Functional Hierarchy

When none of the above organizational schemes prove helpful, the overall functionality can be organized into a hierarchy of functions organized by common inputs, common outputs, or common internal data access. Data flow diagrams and data dictionaries can be used to show the relationships between and among the functions and data.

Additional Comments

Whenever a new Requirements Specification is contemplated, more than one of the organizational techniques given above may be appropriate. In such cases, organize the specific requirements for multiple hierarchies tailored to the specific needs of the system under specification.

There are many notations, methods, and automated support tools available to aid in the documentation of requirements. For the most part, their usefulness is a function of organization. For example, when organizing by mode, finite state machines or state charts may prove helpful; when organizing by object, object-oriented analysis may prove helpful; when organizing by feature, stimulus-response sequences may prove helpful; and when organizing by functional hierarchy, data flow diagrams and data dictionaries may prove helpful.